

ECE 4520/5520 - Homework 2 Solutions

Problem 1.) - Reflection (Internal & External)

1a.)

A ray of light that is traveling in a glass medium of refractive index $n_1 = 1.327$ becomes incident on a less dense glass medium of refractive index $n_2 = 1.117$. Suppose that the free-space wavelength (λ) of the light ray is 1.8 μm . What is the penetration depth of the evanescent wave into medium 2 when the incident angle $\theta_i = 57$ degrees and $\theta_i = 77$ degrees? What is the phase change in the reflected wave when the incident angle $\theta_i = 57$ degrees and $\theta_i = 77$ degrees? What should the minimum incidence angle for TIR be?

1b.)

Also, separately, consider the reflection of light at a normal incidence on a boundary between a glass medium of refractive index $n_1 = 1.23$ and air of refractive index $n_2 = 1$. If the light is traveling from glass to air, what is the reflective coefficient and the intensity of the reflected light? If the light is traveling from air to glass, what is the reflection coefficient and the intensity of the reflected light?

Problem 1a.)

Minimum Incidence Angle For TIR

$$\textcircled{i} \sin(\theta_c) = \left(\frac{n_2}{n_1}\right) ; \sin(\theta_c) = \frac{1.117}{1.327} \approx 0.8417483$$

$$\theta_c = \sin^{-1}\left(\frac{1.117}{1.327}\right) \rightarrow \boxed{\theta_c = 57.3252^\circ}$$

$$\textcircled{ii} \theta_i \approx 57^\circ \dots \theta_i < \theta_c \dots \text{slightly}$$

$$\tan\left(\frac{1}{2} \phi_\perp\right) = \frac{[\sin^2(\theta_i) - n^2]^{1/2}}{\cos(\theta_i)} \rightarrow \frac{[\sin^2(57^\circ) - \left(\frac{1.117}{1.327}\right)^2]^{1/2}}{\cos(57^\circ)}$$

$$\tan\left(\frac{1}{2} \phi_i\right) \approx \frac{(0.7034 - 0.7085)^{1/2}}{0.54464}$$

$$\tan\left(\frac{1}{2} \phi\right) = 0.13112$$

ECE 4520/5520 - Homework 2 Solutions

Problem 1a.) cont.)

$$\tan\left(\frac{1}{2}\phi\right) \approx j0.13112 \quad ; \quad \frac{1}{2}\phi = \tan^{-1}(j0.13112) * 2$$

$$\boxed{\phi_{\perp} = j15.1123^{\circ}}$$

$$\tan\left(\frac{1}{2}\phi_{\parallel} + \frac{1}{2}\pi\right) = \frac{[\sin^2(\theta_i) - n^2]^{\frac{1}{2}}}{n^2 \cos(\theta_i)} = \frac{1}{n^2} \tan\left(\frac{1}{2}\phi_{\perp}\right)$$

$$\tan\left(\frac{1}{2}\phi_{\parallel} + \frac{1}{2}\pi\right) = \left(\frac{n_1}{n_2}\right)^2 \cdot \tan(\phi_{\perp}/2)$$

$$\tan\left(\frac{1}{2}\phi_{\parallel} + \frac{1}{2}\pi\right) = \left(\frac{1.327}{1.117}\right)^2 \cdot \tan\left(\frac{j15.1123}{2}\right)$$

$$\tan\left(\frac{1}{2}\phi_{\parallel} + \frac{1}{2}\pi\right) = j0.1851$$

$$\frac{1}{2}\phi_{\parallel} + \frac{1}{2}\pi = \tan^{-1}(j0.1851)$$

$$\frac{1}{2}[\phi_{\parallel} + \pi] = \frac{\tan^{-1}(j0.1851)}{\frac{1}{2}}$$

$$[\phi_{\parallel} + \pi] = 2 \tan^{-1}(j0.1851)$$

$$\phi_{\parallel} = 2 * \tan^{-1}(j0.1851) - \pi \rightarrow$$

$$\boxed{\phi_{\parallel} = -3.1416 + j21.4582}$$

ECE 4520/5520 - Homework 2 Solutions

Problem 1a.) cont.)

(iii) For $\theta_i = 77^\circ$

$$\tan\left(\frac{1}{2}\phi_\perp\right) = \frac{\left[\sin^2\theta_i - n^2\right]^{1/2}}{\cos(\theta_i)} = \frac{\left[\sin^2(77) - \left(\frac{1.117}{1.327}\right)^2\right]^{1/2}}{\cos(77)}$$

$$\boxed{\phi_\perp = 2 * \arctan(2.1817) \rightarrow \phi_\perp = 130.8^\circ}$$

$$\tan\left(\frac{1}{2}\phi_{//} + \frac{1}{2}\pi\right) = \frac{1}{n^2} \tan\left(\frac{1}{2}\phi_\perp\right)$$

$$\tan\left(\frac{1}{2}\phi_{//} + \frac{1}{2}\pi\right) = \left(\frac{1.327}{1.117}\right)^2 \tan\left(\frac{130.8}{2}\right)$$

$$\frac{1}{2}\phi_{//} + \frac{1}{2}\pi = \tan^{-1}\left[\left(\frac{1.327}{1.117}\right)^2 \tan\left(\frac{130.8}{2}\right)\right]$$

$$\phi_{//} = 2 * \tan^{-1}\left[\left(\frac{1.327}{1.117}\right)^2 \tan\left(\frac{130.8}{2}\right)\right] - \pi$$

$$\boxed{\phi_{//} = 140.9128^\circ}$$

ECE 4520/5520 - Homework 2 Solutions

Problem 1a.) cont.)

(iv) Penetration Depth...

For $\theta_i = 57^\circ$

$$\alpha_2 = \frac{2\pi n_2}{\lambda_0} \left[\left(\frac{n_1}{n_2} \right)^2 \sin^2(\theta_i) - 1 \right]^{1/2}$$

$$\alpha_2 = \frac{2\pi(1.117)}{(1.8 \times 10^{-6})} \left[\left(\frac{1.327}{1.117} \right)^2 \sin^2(57^\circ) - 1 \right]^{1/2}$$

$$\alpha(\text{Depth}) = 1/\alpha_2 \approx -3_j \text{ } \mu\text{m}$$

For $\theta_i = 77^\circ$

$$\alpha_2 = \frac{2\pi n_2}{\lambda_0} \left[\left(\frac{n_1}{n_2} \right)^2 \sin^2(\theta_i) - 1 \right]^{1/2}$$

$$\alpha_2 = \frac{2\pi(1.117)}{1.8 \times 10^{-6}} \left[\left(\frac{1.327}{1.117} \right)^2 \sin^2(77^\circ) - 1 \right]^{1/2}$$

$$\alpha(\text{Depth}) = 1/\alpha_2 \approx 0.4005 \text{ } \mu\text{m}$$

ECE 4520/5520 - Homework 2 Solutions

Problem 1b.)

1b.)

Also, separately, consider the reflection of light at a normal incidence on a boundary between a glass medium of refractive index $n_1 = 1.23$ and air of refractive index $n_2 = 1$. If the light is traveling from glass to air, what is the reflective coefficient and the intensity of the reflected light? If the light is traveling from air to glass, what is the reflection coefficient and the intensity of the reflected light?

$$n_1 = 1.23 \quad ; \quad n_2 = 1$$

① Glass to Air

$$n_1 = 1.23 \quad \dots \quad n_2 = 1$$

$$r_{||} = r_{\perp} = \frac{n_1 - n_2}{n_1 + n_2} \approx \frac{1.23 - 1}{1.23 + 1} = \frac{0.23}{2.23} \approx 0.1031$$

$$r_{||} = 0.1031 \quad \dots \quad \text{positive so no phase shift}$$

$$\text{Reflectance} \approx (0.1031)^2 \rightarrow \text{reflectance} \approx 0.0106$$

② Air to Glass

$$n_1 = 1 \quad ; \quad n_2 = 1.23$$

$$r_{||} = \frac{n_1 - n_2}{n_1 + n_2} = \frac{-0.23}{2.23} \approx -0.1031$$

$$r_{||} = -0.1031 \quad \dots \quad \text{negative} \dots \text{thus, phase shift } 180^\circ$$

$$\text{Reflectance} = (-0.1031)^2 \rightarrow \text{reflectance} \approx 0.0106$$

ECE 4520/5520

homework 2 Solutions

Problem 2.)

Problem 2.) - Absorption, Reflectance, & Reststrahlen Peak

An InP crystal has a refractive index (real part) n of 3.1276 and an extinction coefficient K of 0.279 at a wavelength of 599 nm (photon energy of 2 eV). Calculate the absorption coefficient α of InP at this wavelength and the reflectance of the air-InP crystal surface.

ECE 4520/5520 - Homework 2 Solutions

Problem 2a.)

$$n = 3.1276$$

$$k = 0.279$$

$$\lambda = 599 \text{ nm}$$

photon $e \dots 2 \text{ eV}$

① Absorption Coefficient (α)

$$\alpha = 2k'' \dots = 2k_0 k = 2[(2\pi)(599 \times 10^{-9})](0.279)$$

$$\alpha = 5.8531 \times 10^6 \text{ m}^{-1}$$

② Reflectance (R)

$$\text{Reflectance } (R) = \frac{(n-1)^2 + k^2}{(n+1)^2 + k^2} = \frac{(3.1276-1)^2 + (0.279)^2}{(3.1276+1)^2 + (0.279)^2}$$

$$\text{Reflectance } (R) = \frac{(4.5267) + (0.0778)}{(17.0371) + (0.0778)} = \frac{4.6045}{17.1149}$$

$$\text{Reflectance } (R) = 0.2690 \text{ or } 26.9\%$$

ECE 4520/5520 - Homework 2 Solutions

Problem 3.)

Problem 3.) - Coherence Time & Coherence Length

3a.)

Light emitting diodes (LEDs) emit light that is seemingly of a well-defined color, or wavelength. However, they have much broader spectral widths than lasers. For example, a red LED emitting at 690 nm will have a spectral line width of 28 nm. Find the coherence time and length of the emitted radiation.

3b.)

Also, using the following light sources, find the coherence length

1. A specially designed single mode and stabilized HeNe laser with a spectral width of 150 MHz
2. A quantum well semiconductor laser diode emitting at 1527 nm with a spectral width of 0.05 nm
3. An LED emitting at 1527 nm with a spectral width of 137 nm
4. A multimode HeNe laser with a spectral frequency width of 1.37 GHz
5. Finally, a semiconductor laser diode emitting at 1527 nm with a spectral width of 2.3 nm

Problem 3a.)

$$\lambda = 690 \times 10^{-9} \text{ m} \quad \dots \quad \Delta\lambda = 28 \times 10^{-9} \text{ m}$$

$$\frac{\Delta\nu}{\Delta\lambda} \approx \left| \frac{d\nu}{d\lambda} \right| \approx \left| -\frac{c}{\lambda^2} \right| \quad \dots \quad \Delta\nu = (\Delta\lambda) \left(\frac{c}{\lambda^2} \right)$$

$$\Delta\nu = \frac{(28 \times 10^{-9} \text{ m}) * (3 \times 10^8 \frac{\text{m}}{\text{s}})}{(690 \times 10^{-9} \text{ m})^2} = \Delta\nu = 1.7643 \times 10^{13} \text{ Hz}$$

$$\text{Coherence Time } (\Delta t) = 1 / \Delta\nu \rightarrow 5.668 \times 10^{-14} \text{ seconds}$$

$$\text{Coherence Length } (L) = (c) (\Delta t) \rightarrow 1.7004 \times 10^{-6} \text{ m}$$

ECE 4520/5520 - Homework 2 Solutions

Problem 3b.) Find Coherence Length ...

i

1. A specially designed single mode and stabilized HeNe laser with a spectral width of 150 MHz

HeNe Laser ... $150 \times 10^6 \text{ Hz}$

$$\Delta \nu = \frac{1}{\Delta t} \rightarrow \text{Coherence time } (\Delta t) = 1/\Delta \nu$$

$$\Delta t = \frac{1}{150 \times 10^6} \rightarrow \Delta t = 6.6667 \times 10^{-9} \text{ sec}$$

$$\text{Coherence Length } (l) = c * \Delta t = (3 \times 10^8 \frac{\text{m}}{\text{s}}) * (\Delta t)$$

$$\text{Coherence Length } l = 2.000 \text{ meters}$$

ii

2. A quantum well semiconductor laser diode emitting at 1527 nm with a spectral width of 0.05 nm

Semiconductor Diode ... $\lambda = 1527 \times 10^{-9} \text{ m}$...

$$\Delta \nu = \Delta \lambda (c / \lambda^2) \dots \Delta \nu = (0.05 \times 10^{-9}) (3 \times 10^8 / (1527 \times 10^{-9})^2)$$

$$\Delta \nu = 6.4330 \times 10^9 \text{ Hz}$$

$$\text{Coherence Time } (\Delta t) = 1/\Delta \nu \rightarrow \Delta t = 1.5545 \times 10^{-10} \text{ s}$$

$$\text{Coherence Length } (l) = c * \Delta t \rightarrow (3 \times 10^8 \frac{\text{m}}{\text{s}}) (\Delta t)$$

$$\text{Coherence Length } (l) = 0.0466 \text{ meters}$$

iii

3. An LED emitting at 1527 nm with a spectral width of 137 nm

$$\Delta \nu = (\Delta \lambda) (c / \lambda^2) \rightarrow \Delta \nu = (137 \text{ nm}) [(3 \times 10^8 \frac{\text{m}}{\text{s}}) / (1527 \text{ nm}^2)]$$

$$\Delta \nu = 1.7626 \times 10^{13} \text{ Hz}$$

$$\text{Coherence Time } (\Delta t) = 1/\Delta \nu \rightarrow \Delta t = 5.6734 \times 10^{-14} \text{ sec}$$

$$\text{Coherence Length } (l) = c \Delta t \rightarrow l = 1.7020 \times 10^{-5} \text{ meters}$$

ECE 4520/5520 - Homework 2 Solutions

Problem 3b.) Find Coherence Length...

(iv)

4. A multimode HeNe laser with a spectral frequency width of 1.37 GHz

$$\text{Coherence Time } (\Delta t) = 1/\Delta \nu \approx 7.2993 \times 10^{-10} \text{ sec}$$

$$\text{Coherence Length } (L) = c * \Delta t = 0.2190 \text{ m}$$

(v)

5. Finally, a semiconductor laser diode emitting at 1527 nm with a spectral width of 2.3 nm

$$\Delta \nu = (\Delta \lambda) (c / \lambda^2) \rightarrow \Delta \nu = (2.3 \text{ nm}) ((3 \times 10^8) / (1527 \times 10^{-9})^2)$$

$$\Delta \nu = 2.9592 \times 10^{11} \text{ Hz}$$

$$\text{Coherence Time } (\Delta t) = 1/\Delta \nu = 3.3793 \times 10^{-12} \text{ sec}$$

$$\text{Coherence Length } (L) = c * \Delta t = 0.0010 \text{ meters}$$

ECE 4520/5520 - Homework 2 Solutions

Problem 4.)

Problem 4.) - Fabry-Perot Cavity & Q-Factor

4a.)

Consider a Fabry-Perot optical cavity made of a semiconductor material with mirrors at its ends. (The mirrors have been obtained by coating the end of the semiconductor crystal.) The length of the semiconductor, and hence the cavity, is 237 μm and mirrors at the ends have a reflectance of 0.793. Calculate the cavity mode nearest to the free-space wavelength of 1307 nm. Calculate the separation of the modes, finesse, spectral width of each mode in frequency and wavelength, and the Q-factor.

4b.)

Consider a ruby crystal of diameter 0.7 cm and length 7.4 cm. The refractive index is 1.65. The ends have been silvered and the reflectances are 0.956 and 0.923 each. What is the nearest mode number that corresponds to a radiation of wavelength 635.7 nm? What is the actual wavelength of the mode closest to 635.7 nm? What is the mode separation in frequency and wavelength? What are the finesse F and Q-factors for the cavity?

Problem 4a.)

$$l = 237 \times 10^{-6} \text{ m} ; R = 0.793 ; \lambda = 1307 \times 10^{-9} \text{ m}$$

My chosen index of refraction is $n = 3.6$

$$\text{Mode } (m) = \frac{(2)(n)(L)}{(\lambda)} = \frac{(2)(3.6)(237 \times 10^{-6})}{(1307 \times 10^{-9})}$$

$$\text{mode } (m) = 1.3007 \times 10^3 = 1300.7 \rightarrow 1300$$

$$\text{mode } (m) = \text{Integer}(m) = 1300$$

ECE 4520/5520 - Homework 2 Solutions

Problem 4a.) cont.)

$$\text{Mode Wavelength } (\lambda) = \frac{2nL}{m} = \frac{(2)(3.6)(237 \times 10^{-6})}{(1300)}$$

$$\text{Mode Wavelength } (\lambda) = 1.3077 \times 10^{-6} \approx 1307.7 \text{ nm}$$

$$\text{Mode Separation } (\Delta \nu_m) = \nu_f = \frac{c}{2nL}$$

$$\text{Mode Separation } (\Delta \nu_m) = 1.7581 \times 10^{11} \text{ Hz} = 175.81 \text{ GHz}$$

$$\text{Finesse } (F) \rightarrow F = \frac{\pi R^{1/2}}{1-R} \approx \frac{\pi(0.793)^{1/2}}{1-0.793}$$

$$\text{Finesse } (F) = 13.5150$$

$$\text{Spectral width } \left(\frac{\nu_m}{F}\right) = 1.3009 \times 10^{10} \text{ Hz} \approx 13.009 \text{ GHz}$$

$$\text{Mode Frequency } (\nu_m) = c/\lambda_m \approx 2.2953 \times 10^{14} \text{ Hz}$$

$$\sigma_{\lambda_m} = \sigma\left(\frac{c}{\nu_m}\right) = \left| -\frac{c}{\nu_m^2} \right| \sigma_{\nu_m} = \frac{3 \times 10^8}{(2.2953 \times 10^{14})^2} * 1.3009 \times 10^{10}$$

$$\sigma_{\lambda_m} = 7.4077 \times 10^{-11} \text{ meters}$$

$$\text{Q-Factor} = mF = (1300)(13.5150)$$

$$\text{Q-Factor} = 1.7570 \times 10^4 \approx 17,570$$

ECE 4520/5520 - Homework 2 Solutions

Problem 4b.)

4b.)

Consider a ruby crystal of diameter 0.7 cm and length 7.4 cm. The refractive index is 1.65. The ends have been silvered and the reflectances are 0.956 and 0.923 each. What is the nearest mode number that corresponds to a radiation of wavelength 635.7 nm? What is the actual wavelength of the mode closest to 635.7 nm? What is the mode separation in frequency and wavelength? What are the finesse F and Q -factors for the cavity?

$$d = 0.7 \text{ cm}; l = 7.4 \text{ cm}; n = 1.65; R_1 = 0.956 \\ R_2 = 0.923; \lambda_i = 635.7 \text{ nm}$$

$$M = \frac{2 * L}{\lambda/n} = \frac{(2)(7.4 \text{ cm})}{\frac{635.7 \text{ nm}}{1.65}} \rightarrow m \approx 384570$$

$$\lambda_o = \frac{2Ln}{m_o} \approx \frac{(2)(7.4 \text{ cm})(1.65)}{384570} \rightarrow \lambda_o = 634.99 \text{ nm}$$

$$\text{Frequency Separation: } \Delta \nu_m = \nu_{m+1} - \nu_m = \frac{c}{\lambda_{m+1}} - \frac{c}{\lambda_m}$$

$$\text{Frequency Separation: } \frac{c}{2Ln/(m+1)} - \frac{c}{2Ln/m} = \frac{c}{2Ln}$$

$$\text{Frequency Separation} = \frac{3 \times 10^8}{(2)(7.4 \text{ cm})(1.65)} = 1.2285 \times 10^9 \text{ Hz}$$

$$\text{Frequency Separation} = 1.2285 \times 10^9 \text{ Hz}$$

$$\text{Wavelength Separation} = \Delta \lambda = \frac{\lambda^2}{2Ln} = 1.6512 \times 10^{-12} \text{ m}$$

ECE 4520/5520 - Homework 2 Solutions

Problem 4b.)

$$\text{Geometric Refractive } (R) = (R_1 R_2)^{0.5}$$

$$\text{Geometric Refractive } (R) = [(0.956)(0.923)]^{0.5}$$

$$\text{Geometric Refractive } (R) = 0.9394$$

$$\text{Finesse } (F) = \frac{\pi R^{1/2}}{1-R} = 50.2461$$

$$\text{Finesse } (F) = 50.2461$$

$$\text{Quality Factor} = mF = (384570)(50.2461)$$

$$\text{Quality Factor} = 1.9323 \times 10^7$$